

Investigating the effect of a natural botanical formula on cognitive performance: A randomised double-blind placebo-controlled trial.

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1 Summary

This project tests a novel combination of natural botanical formulae that have pre-existing evidence to aid the maintenance of brain health and cognitive performance for athletes. As a natural and food-based approach, this would have particular relevance for elite athletes who need to avoid World Anti-Doping Agency (WADA) prohibited substances.

This research investigates the use case for the CONKA formula in professional sports to support existing evidence that the combination of natural ingredients can contribute to the maintenance of brain health and the improvement of cognitive performance. All participants of this trial are current professional sports athletes of Harlequins F.C.

It was observed that the cohort administered with the CONKA formula exhibited a marked enhancement in cognitive performance **(+14.86%)** in stark contrast to the placebo control group, which demonstrated a discernible decline in the same metric **(-0.69%)**. The application of a paired t-test to the collected data yielded statistically significant results, a fact that is corroborated by the data presented in Table 3.

The aims of this project were 1) To determine if the CONKA formula has a positive effect on cognitive performance and well-being in elite contact sport athletes and 2) To assess if there is a positive effect of the CONKA formula on the speed of cognitive performance recovery post-fixtures.

The methodology employed for this investigation was a randomised, double-blind, placebo-controlled trial, encompassing a cohort of twenty-nine male professional rugby athletes, dichotomised into two distinct groups. Following a preliminary two-week phase dedicated to baseline testing, 18 participants were randomly assigned to receive the CONKA formula (intervention), whereas 11 participants were similarly allocated to a placebo regimen, over a duration of four weeks.

The protocol mandated that participants undergo cognitive assessments at intervals of two, four, and six days subsequent to matches scheduled every six or seven days, throughout a six-week trial period. This period included the initial two-week baseline testing phase preceding the administration of either the placebo or the natural daily formula.

The evaluation of their information processing capability was conducted through the Integrated Cognitive Assessment (ICA). Additionally, memory and focus were assessed utilising a digitised version of the Sports Assessment Concussion Test 6th Edition (SCAT 6).

Contact exposure was measured using integrated mouthguards recording peak linear acceleration measured in gravitational force (G-force), and peak rotational forces (radians per second squared, RADS). Athletes' match minutes, preexisting supplement use and daily wellness questionnaires were used to record soreness, stress, readiness, sleep duration, sleep satisfaction and extra recovery completed.

2 Background

Cognitive impairment imposes personal, economic, and societal burdens, and thus the proactive management of cognitive performance in sport not only presents an opportunity for advancements in well-being but also financial, and competitive advantages.

In rugby, repeated sub-concussive head traumas cause a loss of cognitive performance and blood flow to the brain after just one season resulting in a higher risk of long-term neurodegeneration⁵. In Soccer, 80% of athletes are considered cognitively impaired after just 20 throw-in headers with a modern synthetic ball⁶⁷. Furthermore, cognitive impairment has been associated with a 60% risk of subsequent injury (of any type) in rugby union¹ and 50% in association with football (soccer)². In addition, declines in cognitive performance have been shown to increase the risk of sustaining lower limb injuries in college-level contact sports³ and subsequent injuries also occur more quickly following a concussion than a non-concussive injury. In instances where an individual has sustained two traumatic brain injuries (TBIs) there has been a 33% rise in the risk of dementia and for five or more TBIs, the risk rises by 200%⁴. Cognitive performance management among athletes, therefore, not only provides an opportunity for improvements in well-being but an opportunity for the reduction in personal, societal and economic burdens associated with loss of cognitive performance.

Over the last few decades, certain nootropic and adaptogenic compounds have indicated potential for the management of cognitive performance. Observational and intervention studies have also demonstrated associations between cognition and diet, including omega-3 fatty acids, antioxidants, herbs, and plant-based flavonoids with mechanisms attributed to neurotransmission, neuroprotection, neuroplasticity, cognitive enhancement, vasodilation/analgesia, and neurorepair.

In summary, there is an evidence base to suggest that biologic (food-based) nootropics and adaptogens offer substantial potential for the maintenance of brain health and subsequent improvements in cognitive performance.

2.1 Research Aims

1. To understand whether the natural botanical formula (intervention) positively impacts the cognitive performance and well-being of athletes.
2. To understand whether the intervention positively impacts the speed of cognitive performance recovery post-fixture in athletes.

3 Methods

This study will investigate the impact of the intervention formulas' ability to support cognitive performance and the maintenance of brain health among professional athletes over 6 weeks, including a 2-week baseline testing period and 4 weeks of intervention. The intervention is an adaptogenic and nootropic blend of six well-established naturally occurring ingredients. All six components in Table 6 have

a strong evidence base and have been used frequently in human research trials. The placebo is a taste and visual mimicking compound of the intervention formula, containing a combination of glycerin, honey and citric acid which the intervention formula does not contain.

3.1 Participants

A total of 29 participants completed this study (Male $n = 29$). Participant characteristics are presented in Table 1. No participants withdrew from the trial.

Table 1: Participant Characteristics

Characteristic	Total (n=29)	Control (n=11)	Intervention (n=18)
Age (Yr.)	25.87 ± 4.34	25.08 ± 3.11	24.76 ± 5.56
Height (m)	1.78 ± 0.09	1.7 ± 0.11	1.817 ± 0.08
Mass (kg)	100.20 ± 11.66	96.83 ± 13.80	103.01 ± 13.15
Type	Backs: 14, Forwards: 16	Backs: 5, Forwards: 7	Backs: 7, Forwards: 9

Data is presented as mean $\pm$ standard deviation. Yr., Years; m, Metres; kg, Kilograms; Mins, game time; n, number of participants.

Table 2: Number of participants with IMG, match minutes breakdown by group & additional supplementation

	Intervention	Control
Match Play (mins)	244.93 ± 84.48	174 ± 137.05
IMG	n=10	n=7
Supplement use	n=10	n=5
Omega 3	n=7	n=1
Creatine	n=9	n=3
Creatine & Omega 3	n=7	n=1
Multivitamin	n=6	n=4
Turmeric	n=2	n=0

IMG, Integrated Mouth Guard

Table 3-5: Compliance in 14-day testing intervals (visualised in appendix 2)

3.2 Compliance

Compliance refers to the cognitive tests that were completed on the correct pre-designated days of the week. Scores are sectioned into equal 14-day periods to be directly comparable to the baseline period 1 (p1).

Table 3: Compliance in 14-day Testing Intervals - Period 1 06.11.23 - 20.11.23 (P1)

Day	Control Group P1		Intervention Group P1	
	Actual	Expected	Actual	Expected
Day 2	106	33	187	54
Day 4	75	33	119	54
Day 6	57	33	83	54

Table 4: Compliance in 14-day Testing Intervals - Period 2 21.11.23 - 07.12.23 (P2)

Day	Control Group P2		Intervention Group P2	
	Actual	Expected	Actual	Expected
Day 2	16	22	24	36
Day 4	31	33	57	54
Day 6	27	33	45	54

Table 5: Compliance in 14-day Testing Intervals - Period (P3)

Day	Control Group P3		Intervention Group P3	
	Actual	Expected	Actual	Expected
Day 2	17	33	49	54
Day 4	27	33	23	54
Day 6	5	22	13	36

3.3 Intervention

The intervention comprised the ingredients listed in Table 6. The intervention is a natural formula and was provided as a liquid. The total dose was divided into two servings for consumption in the morning and in the evening.

Possible contra-indications to the supplement are listed in Table 7. Participants were informed of the ingredients within the study and all participants signed a standard consent form regarding the ingredients and how the data would be analysed (see appendix 1).

Table 6: Constituents of the product

Common Name	Scientific Name/Details
Ashwagandha root	Withania somnifera
Turmeric	Curcuma longa with black pepper (5%)
Lemon balm	Melissa officinalis
Rhodiola rosea root	Rhodiola rosea
Bilberry fruit	Vaccinium myrtillus

Table 7: Known possible, but rare, side effects at high doses of constituents

Herb	Contra-indications
Ashwagandha root	Pregnancy, breastfeeding, sedatives, thyroid meds, diabetic agents
Turmeric	Pregnancy, breastfeeding at therapeutic dose. Anti-platelet, anti-coagulant drugs at high doses
Lemon balm	Hypothyroid effects
Rhodiola rosea	Pregnancy, breastfeeding. Caution with anti-diabetic drugs and anti-hypertensive drugs
Bilberry	None known though high doses anthcyanosides can inhibit platelet aggregation

3.4 Assessments

3.4.1 CognICA

CognICA is a computerised cognitive test, based on rapid categorisation task, and is independent of language, education, culture and learning effects. CognICA has been shown to accurately detect mild cognitive impairment, is able to measure small changes in cognitive function and is moderately to highly correlated with other popular cognitive tests.

3.4.2 Digitalised Sports Assessment Concussion Tool (SCAT 6)

The SCAT6 is a standardised diagnosis tool for suspected head injuries in sports. The CONKA software version 2.2.5 utilises two key elements of the tool, concentration (focus) and memory sections.

3.4.3 Harlequins Pre-Training Wellness Questionnaire

Each participant answered questions on stress levels, readiness, sleep duration and satisfaction before training. Part of the questionnaire also included levels of muscle soreness present in the lower body, upper body, and neck.

3.5 Statistical analysis

Paired t-tests and ANCOVA tests were performed to determine any statistically significant changes in cognitive performance from baseline tests on days 2, 4, and

6 days post-fixture to trial period days 2, 4, and 6 days post-fixture. An alpha of 0.05 was set for statistical significance. All statistical analysis was conducted in SPSS software (version 29, Statistical Product and Service Solutions, International Business Machines).

4. Results

4.1 Integrated Cognitive Assessment

T-Test

Results from the Integrated Cognitive Assessment Accuracy score alone showed that, on days 4 and 6 post-fixture, the intervention group's scores increased by an average of 5%, while the control group's score change was an average of 1%. A paired t-test comparing baseline and trial period scores yielded a p-value of 0.0018, indicating statistical significance at the alpha level of 0.05. Data from 3 participants (nos. 01, 09) were incomplete and excluded from the t-test analysis.

Table 8: Intervention Group Paired T-Test Results

Participant Number	Baseline	Test Period	Difference	P Value
27	82.5	93.33	10.83	0.0018
26	79.00	83.00	4.00	
24	92.00	97.7	5.67	
21	74.50	77.00	2.50	
18	82.00	86.66	4.66	
17	82.00	82.00	0.00	
16	55.17	60.20	5.06	
15	79.50	81.50	2.00	
13	83.50	88.67	5.17	
12	66.00	67.75	1.75	
29	79.00	87.00	8.00	
28	88.00	96.00	8.00	
7	90.00	92.67	2.67	
6	81.00	80.00	-1.00	
4	93.00	91.33	-1.67	
Overall P Value:				0.0018

ANCOVA

A repeated measures analysis of covariance (ANCOVA) was utilised to evaluate the uniformity of significance levels among participants. The analysis yielded a p-value of 0.09 for the effect of the intervention, so while the intervention contributes to the variance observed, it falls short of achieve statistical significance by 0.04. The outcome implies that the positive performance of the intervention varies too much throughout the sample. Notably, an increase in compliance during the later stages of the trial (Table 3), suggests that increased adherence and a more even distribution of contact exposure (Table 11), across the two groups may resulted in greater statistical significance for the effect of the intervention.

Table 9: Between-Subjects Contrasts

Source	Formula	Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Control	Linear	120.61	1	120.61	8.16	0.009	0.25
Intervention	Linear	46.13	1	46.13	3.12	0.091	0.11

4.2 Digitalised SCAT 6

The most notable difference in the cognitive performance between the two groups was shown on day 4 post-game for memory scores. An average improvement of +10.02% for the intervention group and a decline of -0.96% for the control group. On Day 6 the average focus score improvement was +7.64% for the intervention group compared to the control group that declined by -0.25%.

4.3 Memory

The memory test performance in the intervention group improved by 4.22% whereas the memory performance of the control group reported -1.05% change overall.

4.4 Focus

The focus test performance of the intervention group improved by +5.23% whereas the control group had a 0.41% change overall.

5 Discussion:

Overall, the intervention has a significant positive impact on cognitive performance over a 4 week period. The intervention group improved by +14.86%, including an +5.16% average improvement in the FDA approved accuracy of processing information metric. The control group, by contrast, remained largely unchanged by the placebo with a small decline of -0.69% overall.

In the study, 10 athletes from the intervention group and 7 from the control group utilised smart gum shields to track their exposure to physical contact. Analysis revealed that, on average, athletes in the intervention group experienced 60% more contact exposure compared to their counterparts in the control group, across three key metrics: Workload, Events above 8G, and Accumulated G (see Table 11).

The available data suggests that the intervention group not only showed greater cognitive performance improvements but also endured a significantly higher level of contact exposure throughout the trial period.

This pattern implies that the intervention may have played a role in enhancing the athletes' recovery speed from physical contact, as evidenced by their improved cognitive performance relative to their baseline recovery rates and compared to the control group's recovery throughout the trial. This observation suggests a potential benefit of the intervention in supporting faster cognitive recovery between fixtures.

Within the intervention group of 18 athletes, 5 athletes were concurrently consuming additional supplements, specifically creatine and omega-3 fatty acids, during both the 2-week baseline and the 4-week intervention phases of the 6-week study. The results are detailed in Table 12.

Table 10: *Summary of contact exposure (measured in G force) in 14 intervals over the 6-week total trial period*

	B-line Total (P1)	AveP2	Ave P3	Total	Ave Total Per Athlete
Intervention					
Total Workload	1257.8	1252.8	1871.50	3124.30	312.43
Events Above 8G	306	330	486.20	816.20	81.62
Accumulated G	5415.2	5607	9110.1	14717.1	1471.71
Control					
Total Workload	600.5	490	817.7	1307.7	186.81
Events Above 8G	135	141	159	300	42.85
Accumulated G	2533.5	2353.3	2912.8	5266.1	752.3

b-line,baseline, ave, average, p2, period 2, p3 period 3.

This sub group experienced a 13.32% overall average improvement including a +3.18% improvement in accuracy information processing test. It is therefore not possible to conclude that the use of Creatine and Omega3 supplementation alongside the intervention formula has any positive synergistic effects.

Table 11: A Summary of Total Cognitive Performance Improvements of the Participants Belonging To The Sub-Group Using Additional Supplementation

Participant No.	Match Mins	Total Increase	Processing
29	148	36.44%	9.20%
7	405	5.09%	2.88%
27	166	27.52%	11.61%
15	-	-23.59%	2.45%
4	230	-1.57%	-1.82%

The compliance data in tables 3-5 reveals a distinct trend: both groups show high compliance at the beginning of the study during the initial 14-day baseline period (P1), suggesting that the initial tasks are more manageable or that participants are more motivated at the start, leading to significant over-compliance. However, as the study progresses into phases P2 and, more notably, P3 (the final 14 days), compliance decreases, indicating challenges in maintaining the required levels of participation. This trend of declining compliance becomes especially pronounced in P3, with both groups showing substantial under-compliance, particularly on Day 6, where the control group's compliance dropped by 77.27%, and the intervention group's by 63.89%. This under-compliance on Day 6, the highest for both groups during P3, suggests that participants may lose interest in repeated testing, affecting their ability to comply.

Moreover, 7 participants in the intervention group had notably stronger testing compliance in the final 2 weeks of the trial period. Isolating these scores from the total intervention group reveals a +3% increase on on the the total intervention group average improvement. One possible reason for this could be, as hypothesised, the intervention formula had the maximum amount of time to have an effect and therefore a further increase in cognitive performance was detectable. In order to

discern whether this is a repeatable result it is recommended that further research utilises longer trial duration and puts measures in place to improve compliance at the latter stages of the testing period.

These compliance observations suggest that the intervention may still produce statistically significant results with a more restricted intervention schedule. To address the compliance issues observed, a repeat trial should be designed accordingly, for example, a supervised and standardised cognitive screening location at the training facility or club to be used once weekly as part of a pre-existing weekly screening procedure may ensure higher compliance in the later stages. Additionally introducing more sophisticated methods of brain-tracking technology, such as eye tracking, would require fewer participants to be involved and therefore a more attentive approach from the researchers to the participants could be achieved.

6. Appendices

Appendix 1: Participant Consent Form

Project title: Effect of a CONKA V23 Elite on the cognitive performance and well being of professional athletes

Project title: Effect of a CONKA V23 Elite on the Cognitive and Physical Performance of Professional Athletes

This form is to confirm that you understand the purposes of the project, what is involved and that you are happy to take part. Please tick the box to indicate your agreement.

I confirm that the Harlequins F.C study coordinator has explained the purpose of the trial.

I have had sufficient time to consider the information, and I am satisfied with the answers I have been given.

I understand who will have access to personal data provided, how the data will be stored and what will happen to the data at the end of the project.

I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.

I understand that anonymised (i.e., not identifiable) versions of my data may be archived and shared with others for legitimate research purposes.

I understand that should I experience any suspected adverse side-effects, I should contact the team as soon as possible and stop taking the supplement immediately. If I require medical assistance, I will in the first instance contact 999/111 before the research team.

I confirm that for the duration of this 12-week study I will not start taking any new supplement(s).

I agree to notify the study coordinator should any change in a pre-diagnosed condition occur during the 6-week study.

I confirm that I am not allergic to any of the ingredients included in this study.

I understand that the ingredients used in this study have been batch-tested by Informed Sport (LGC Group).

I agree to take part in the above project.

Name of Participant: _____

Date: _____ / _____ / _____

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